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ICS 2203 WEB APPLICATION DEVELOPMENT 1

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This presentation is intended to covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green Exercise tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

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LESSON 4

The DNS

Learning outcomes

Upon completing this topic, you should be able to:

1. What a DNS is.
2. List three common DNS Types.
3. Advice on the process of registering a Domain Name.

4.1. Introduction

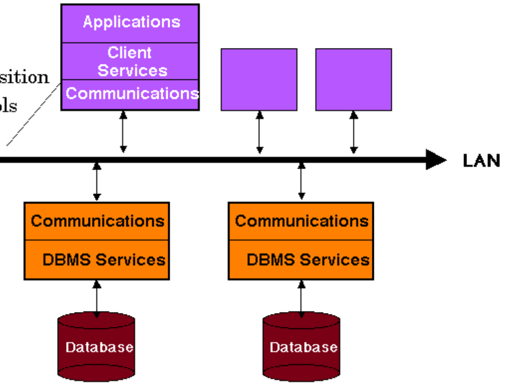
What is a Distributed Database?

- A distributed database (DDB) is a collection of multiple, logically interrelated databases distributed over a computer network.

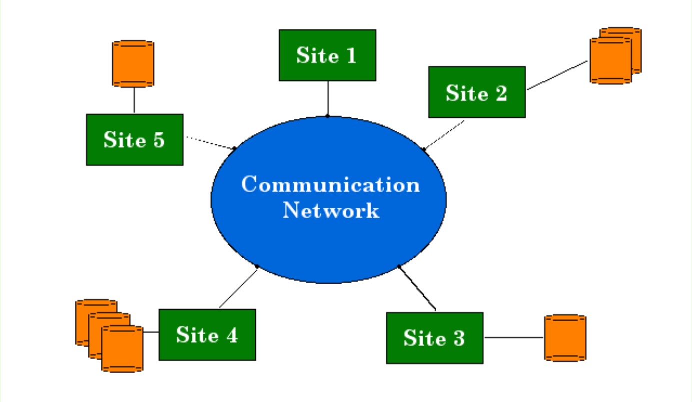
- A distributed database management system (D–DBMS) is the software that manages the DDB and provides an access mechanism that makes this distribution transparent to the users.

- Distributed database system (DDBS) = DDB + D–DBMS

- directory
- caching
- query decomposition
- commit protocols



•.1. **The Figure below shows a DDBMS environment**



What Is a DNS? DNS stands for Domain Name System.

This DNS is a Domain Name System. This system is much similar to the telephone directory system and it allows users to get resources over the internet. A domain name is a unique name for a web site, like google.com. Domain names must be registered. When domain names are registered they are added to a large domain name register, and information about your site - including your internet IP address - is stored on a DNS server. The DNS is a distributed database system that stores domain names of computers alongside their numerical IP addresses. Since users mostly remember names as compared to IP addresses, the system is useful to providing the actual address where an internet resource is stored and this is then used to get and return the resource to the computer requesting for it. It is designed in such a way that no Single database in the world stores all the computer names and addresses. Ideally we have different sites in the world that have these servers and the servers store only a subset of the addresses.

A DNS server is responsible for informing all other computers on the Internet about your domain name and your site address. The DNS architecture is a hierarchical distributed database and an associated set of protocols that define:

- A mechanism for querying and updating the database.
- A mechanism for replicating the information in the database among servers.
- A schema of the database.

4.1.1. Some History

DNS originated in the early days of the Internet when the Internet was a small network established by the United States Department of Defense for research purposes. The host names of the computers in this network were managed through the use of a single HOSTS file located on a centrally administered server. Each site that needed to resolve host names on the network downloaded this file. As the number of hosts on the Internet grew, the traffic generated by the update process, as well as the size of the HOSTS file, increased. The need for a new system, which would offer features such as scalability, decentralized administration, support for various data types, became more and more obvious.

4.1.2. The DNS and the Internet

The internet works in such a way that every domain has to be registered by a domain name registration company. After registration this name and the IP address of the host are stored in a DNS server. The host will be responsible for broadcasting the registered name and IP to other DNS servers.

No one DNS server holds a table of all the registered domains around the world. Hence resolution of these names takes a hierarchical approach starting from the branches coming to the root.

The DNS is a good example of a distributed Kind of a database. The database stores the registered domain names in different zones.

4.1.3. The DNS Functionality

- Given

  -Name of a computer

- -Returns

  - Computer's internet address

- Method
  - Distributed lookup
  - Client contacts server(s) as necessary

Each entry in server consists of, Domain name, DNS type for name, Value to which name corresponds

During lookup, client must supply; Name, Type

The Server Matches both name and type

Example. Examples of DNS Types

- Type A (Address)
  - Value is IP address for named computer
- Type MX (Mail eXchanger)
  - Value is IP address of computer with mail server for name; matches the computer name found in a n email address to an IP address.

- Type CNAME (Computer NAME)
  - Value is another domain name
  - Used to establish alias (www)

4.1.4. The DNS Syntax

- Alphanumeric segments separated by dots

Example ✎. Provide an examples of different Domain Names

1. www.netbook.cs.purdue.edu
2. www.eg.bucknell.edu

Example ✎. Most significant part on right

4.1.5. Obtaining a Domain Name

- Organization
  - Chooses a desired name
  - Must be unique
  - Registers with central authority
  - Placed under one top-level domain
- Names subject to international law for
  - Trademarks
  - Copyright

Top Level Domain Names

Domain Name	Assigned To
com	Commercial organization
edu	Educational institution
gov	Government organization
mil	Military group
net	Major network support center
org	Organization other than those above
arpa	Temporary ARPA domain (still used)
int	International organization
country code	A country

4.1.6. The DNS Client server Interaction

Resolution – this is the translation of a domain name into an equivalent IP address. The software that performs resolution is known as resolver. Each resolver is configured with an address of a local domain server. Multiple DNS servers arranged in hierarchy are used. Each server corresponds to contiguous part of naming hierarchy.

Example ✎. The internet is a System of Systems. Discuss

Solution: Points-Network of networks (MAN s, WANs, LANs etc), Different platforms of computers, Different mediums, Many standards of communication, Many resources to be shared, Dynamic uncontrolled growth

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Revision Questions

Exercise 1. ✍ List any five top level domain names together with their meaning.

Exercise 2. ✍ A DNS server consists of a

1. A domain name
2. A DNS type for name
3. And a value to which the name corresponds.

During look up a client application must supply both the name and the type. Illustrate the three DNS types.

Exercise 3. ✍ Describe the significance of the following terms to the internet

1. Browser
2. Server

3. nsp
4. IP
5. DNS

Exercise 4. ✍ Giving examples of at least two describe a Domain Name registration Company.

References and Additional Reading Materials

- George Beekman,(2008) Tomorrow's Technology and You, Complete, 9/E,Oregon State University Ben Beekman, ISBN-10: 0135045045 • ISBN-13: 9780135045046 ©2010 • Prentice Hall • Paper, 672 pp
- Chuck Musciano and Bill Kennedy(2000)HTML & XHTML: The Definitive Guide, 4th Edition, Oreilly, ISBN 0-596-00026-X

- Craig Grannell(2007 The Essential Guide to CSS and HTML Web Design, Springer, ISBN-13 (pbk): 978-1-59059-907-5

Solutions to Exercises

Exercise 1.

Com commercial

Edu educational

Mil military

Net network

Org organizational

Gov government

Country code a country

Exercise 1

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Exercise 2.

Type A (address) value is the IP of named Computer

Type MX (mail Exchange) value is the IP address of computer with mail exchange for name

Type CName (Computer Name) value is another domain name

Exercise 2

Exercise 3.

1. Browser: - application to access network resources

2. Server – provides services to network clients

3. nsp: - network service provider that sel bandwidth to regional ISP

4. IP: Internet addressing system that is used to retrieve web resourced from hosts

5. DNS: - systems of databases distributed across different sites on the web

Exercise 3

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